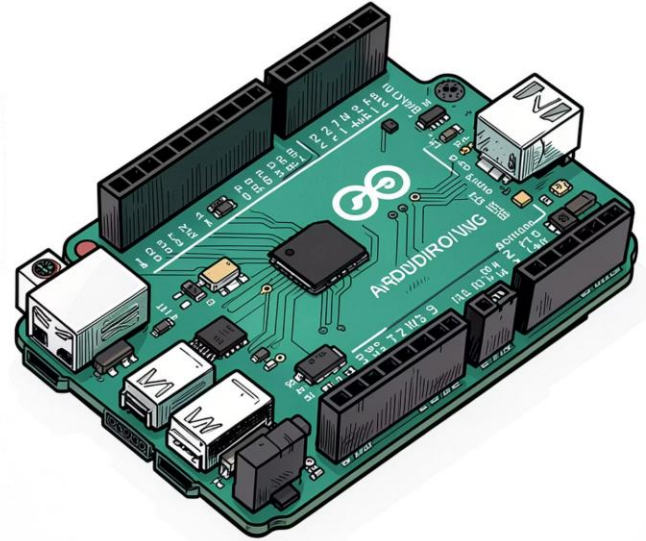


Gas Detector Sensor

By: Anushka Hankare • **School:** DMS School

Modern STEM project: low-cost gas detection using Arduino UNO + MQ-135

- Speaker notes: Welcome the audience briefly — introduce yourself and the project goal: a simple, affordable gas detector for safety demonstrations.



Introduction & Problem Statement



Why gas detection matters

- Undetected gas leaks can cause fires, explosions and long-term health risks
- Common household and laboratory hazards: LPG, smoke, VOCs
- Professional detectors are expensive — schools need affordable options

i Speaker notes: Emphasize real-world incidents and how early detection prevents accidents. Keep examples relatable (kitchen, lab, classroom).

Project Objectives



Detect harmful gases

Continuously monitor air quality for hazardous levels.



Provide instant alert

Activate buzzer when threshold is crossed for immediate warning.



Improve safety

Easy to deploy in homes, schools and small labs.



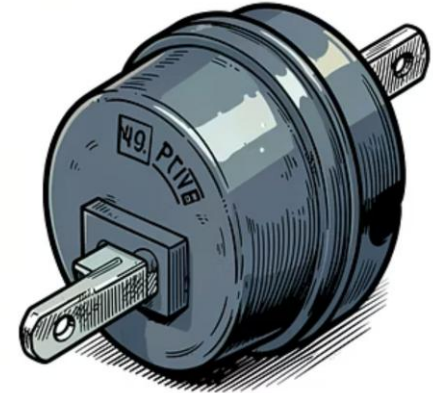
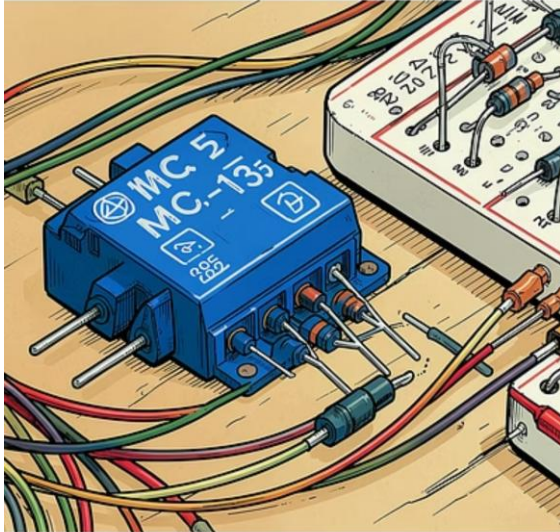
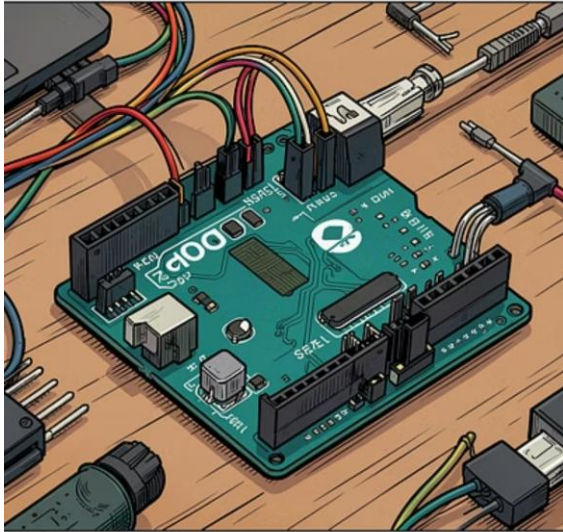
Affordable & educational

Build with Arduino UNO to learn sensor interfacing and programming.

- ☐ Speaker notes: Briefly read each objective; highlight how the project balances practicality and learning outcomes for students.

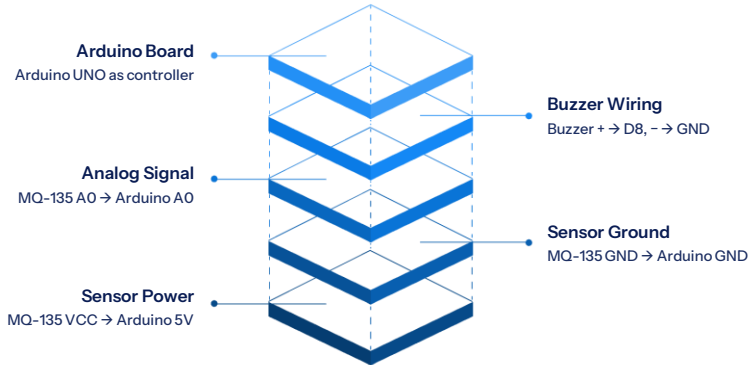
Components Used

Component	Purpose / Notes
Arduino UNO	Microcontroller — reads sensor, controls buzzer
MQ-135 Gas Sensor	Senses air quality / VOCs; analog output (A0)
Active Buzzer	Audible alarm when digital pin drives HIGH
Breadboard	Prototyping and neat wiring
Jumper Wires & USB Cable	Connections and power / programming via USB

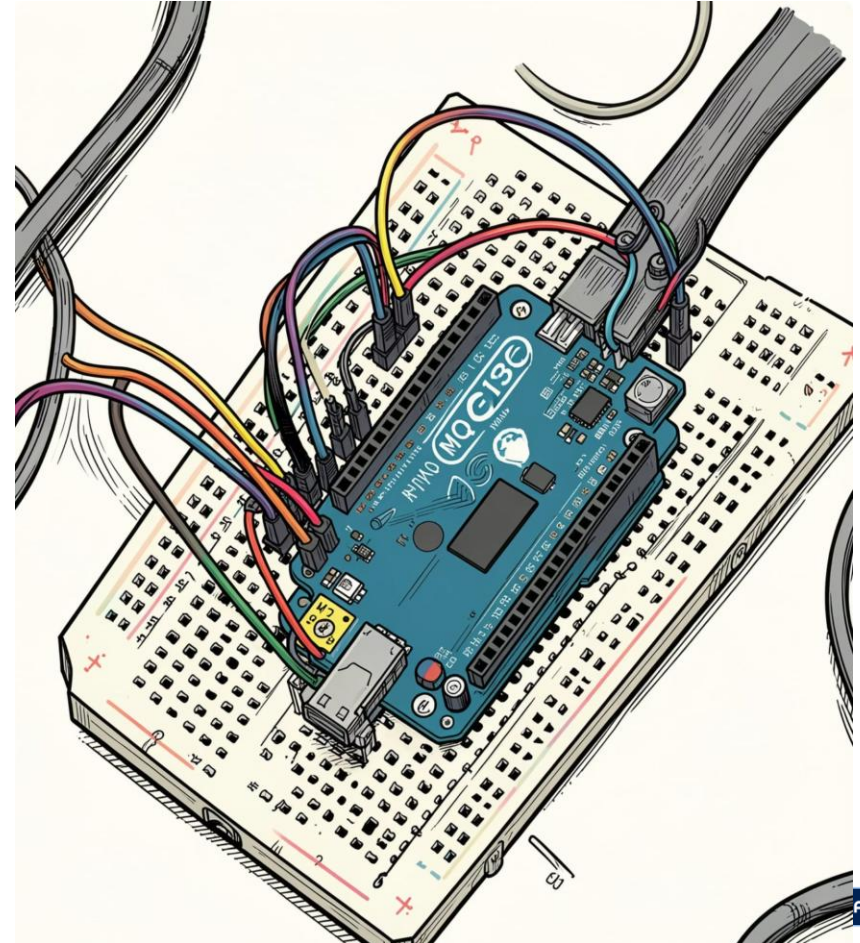


Speaker notes: Show each physical component to the audience; pass around if allowed. Mention that MQ-135 uses only A0 for this demo.

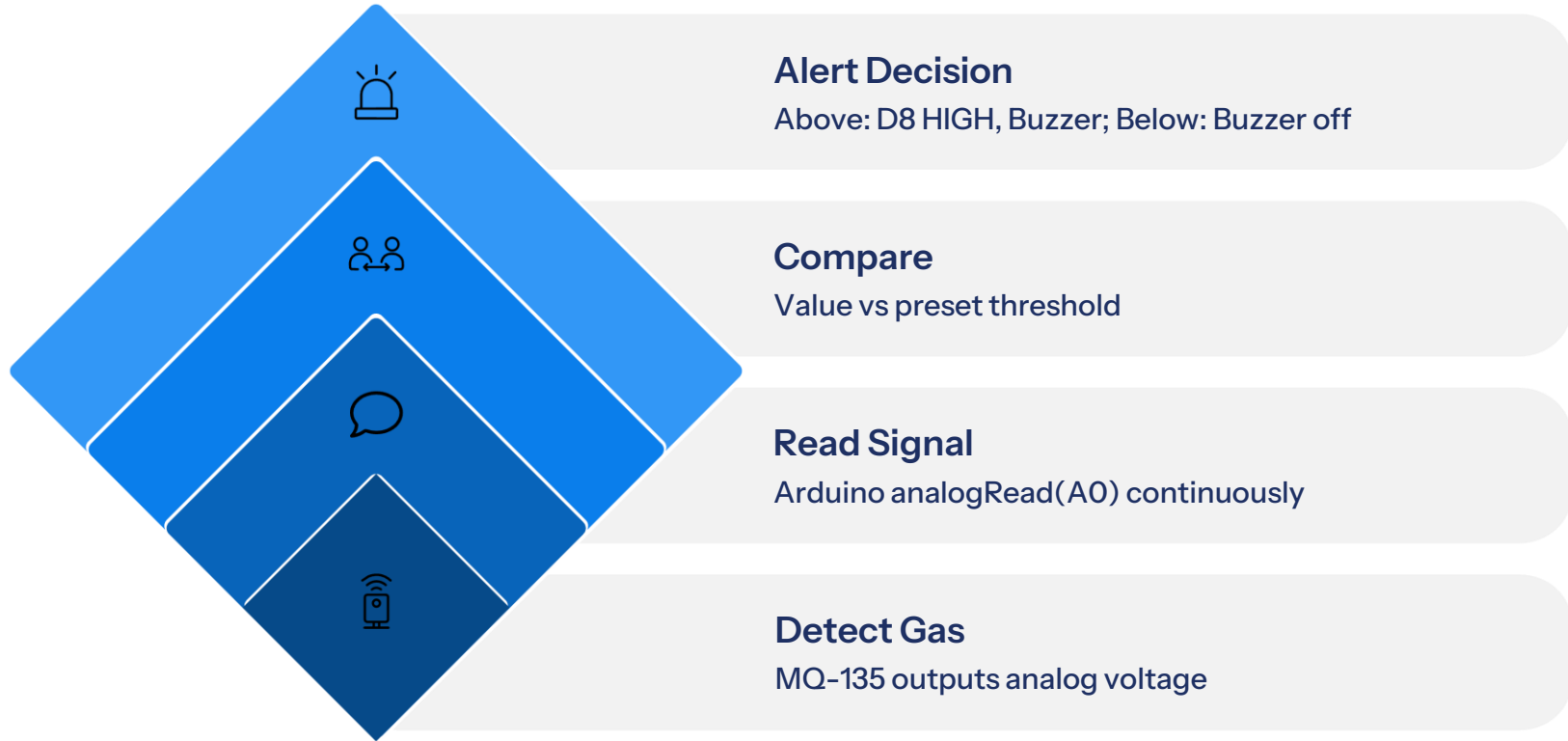
Circuit Diagram & Connections



Block diagram: Sensor → Arduino (A0) → Threshold check → Buzzer (D8)



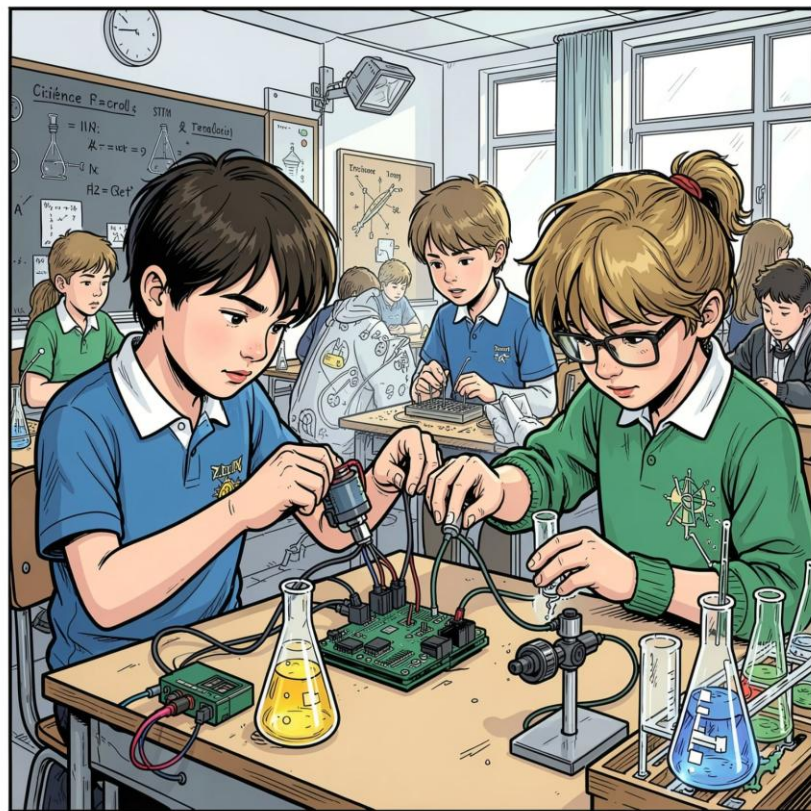
Working Principle



Summary steps:

1. MQ-135 detects gas concentration and outputs analog voltage.
2. Arduino reads value via `analogRead(A0)` continuously.
3. Code compares value to a preset threshold.
4. When exceeded, Arduino sets D8 HIGH; buzzer sounds until level returns to normal.

Results & Testing



Testing notes: Verify with controlled vapours, adjust threshold, ensure stable readings before final demo.

Normal Air
Buzzer: OFF — sensor values below threshold

Harmful Gas Detected
Buzzer: ON — immediate audible alert

Advantages & Applications



Low cost

Affordable parts make it accessible for schools.



Easy to build

Simple wiring on a breadboard; beginner-friendly code.



Fast response

Immediate buzzer activation when threshold crossed.

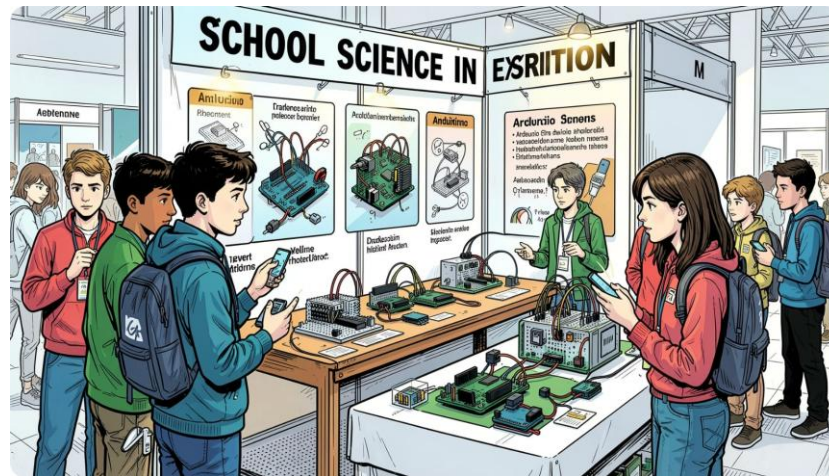


Portable

Can run from USB power or small battery pack.

Applications

- Homes & kitchens — early leak detection
- Schools & laboratories — safety demonstrations
- Workshops & small industries — local VOC monitoring
- Portable safety kits for beginners



Speaker notes: Emphasize versatility — from demonstrations to practical small-area monitoring. Suggest next steps: add LED indicators, data logging, or GSM alerts for expansion.

Conclusion & Thank You

The Gas Detector Sensor successfully uses an MQ-135 and Arduino UNO to detect harmful gases and trigger an instant buzzer alert. Simple, affordable, and effective for classrooms and homes.

Author: Anushka Hankare • **School:** DMS School

■ CONTACT

■ QUESTIONS

- Speaker notes: Close by thanking the audience, invite questions, and offer to demonstrate the device after the talk. Mention resources for further learning and possible improvements (calibration, visual alerts, data logging).

